

CityTech TTP Summer Bootcamp

Intro to React, HTML, and CSS

2026-06-08

Agenda

1. Review
2. Follow Up: Issues and PRs
3. Fishbowl Stand-up
4. HTML
5. React

Review

What are the most important things you remember from last week?

Follow Up: Issues and PRs

On the **main course repo**:

- Keep using **GitHub issues** to ask content questions - be **specific**
- Try to figure it out yourself first; use issues as a **secondary** resource
- Someone on the instructional team will answer as soon as we can
- Want to improve the materials? Open a **pull request** - keep PRs **small and specific** in scope

Fishbowl Stand-up

The instructional team will **model a stand-up** while you watch.

- Each person: what I did **yesterday**, what I'm doing **today**, any **blockers**
- Notice how short and focused it stays

Websites and Web Apps

A **website** presents content; a **web app** is interactive software running in the browser. Both are built from the same core technologies:

- **HTML**: the structure and content
- **CSS**: the styling and layout
- **JavaScript**: the behavior and interactivity

HTML and CSS are the foundation: even React ultimately produces HTML and CSS for the browser to render.

HTML - A Markup Language

HTML = HyperText Markup Language.

- A **markup language** wraps content in **tags** that describe its structure and meaning - a heading, a paragraph, a list, an image
- It **describes** content; it doesn't compute (no logic or loops like a programming language)
- The browser reads your HTML and builds the page from it

A First Look at HTML

```
<article>
  <h1>Chocolate Chip Cookies</h1>
  
  <p>Ready in 30 minutes – makes about two dozen.</p>
  <a href="/recipes">More recipes</a>
</article>
```

- Nested tags are **indented** to show the structure
- `<img />` is a **self-closing tag** - no separate closing tag needed
- Tags take **attributes** that configure them - e.g. `<img>` 's `src` (the image file) and `alt` (a text description for screen readers, or if it fails to load)
- Whether an attribute is **required or optional** depends on the tag

Official HTML References

- **MDN Web Docs - HTML** (the go-to reference)
<https://developer.mozilla.org/en-US/docs/Web/HTML>
- **web.dev - Learn HTML** (Google)
<https://web.dev/learn/html>
- **WHATWG HTML Living Standard** (the official spec)
<https://html.spec.whatwg.org/>

Code Together: Dummy HTML Chat App

Let's hand-write a static chat screen in plain **HTML** - no CSS or JavaScript yet.

- A heading with the chat room's name
- A list of messages (hardcoded for now)
- A text input and a **Send** button

"Dummy" = it's just structure; nothing actually sends yet.

Why React Emerged

- Early web pages were **static**; making them interactive meant hand-editing the DOM, step by step
- As **SPAs** grew, keeping the screen in sync with the data by hand became messy and bug-prone
- **Facebook open-sourced React in 2013** to manage complex, constantly-updating UIs (news feed, chat, notifications)
- Today it's one of the most widely used ways to build web interfaces

What Is React?

- A **JavaScript library** for building user interfaces
- **Components**: reusable, self-contained pieces of UI
- **Declarative**: you describe what the UI should look like; React updates the page
- **JSX**: write HTML-like markup right inside JavaScript

What Is a Function?

In programming, a **function** takes some **input**, does work, and returns an **output**.

- Same input → same output - predictable
- **Reusable**: call it again with different inputs instead of repeating yourself

```
function greet(name) {  
  return "Hello, " + name;  
}  
greet("Maya"); // "Hello, Maya"
```

The Big Idea: $UI = f(state)$

A component is a **function of its data** - give it data, it returns UI. Change the data and React re-renders just that component.

```
function MessageList({ messages }) {  
  return (  
    <ul>  
      {messages.map((m) => <li key={m.id}>{m.text}</li>)}  
    </ul>  
  );  
}
```

- **Static HTML:** you'd hand-write and edit every `<li>` yourself
- **A function:** the list is generated from `messages` - no duplication

React References

- **React docs (home):** <https://react.dev>
- **Learn React** (guided tutorial) - <https://react.dev/learn>
- **API reference:** <https://react.dev/reference/react>

Real-World React Frontends

Open-source React apps worth exploring:

- **Excalidraw**: virtual whiteboard, client-side React + TypeScript
<https://github.com/excalidraw/excalidraw>
- **tldraw**: infinite canvas and drawing
<https://github.com/tldraw/tldraw>
- **TodoMVC**: the classic to-do app, small and readable
<https://github.com/tastejs/todomvc>

What Is Boilerplate?

Boilerplate is the standard, repeated setup code and config every project needs *before* you write any features - folder structure, build config, linting, base dependencies.

- The same scaffolding shows up across countless projects
- Teams start from **starter templates / generators** instead of reinventing it
- Saves time and enforces consistency across a codebase
- Trade-off: it can include things you don't need - know what it sets up

Start a React + TypeScript Project

We use **Vite** (modern and fast) with the React + TypeScript template - and **Yarn**, not npm.

```
yarn create vite my-app --template react-ts
cd my-app
yarn          # install dependencies
yarn dev      # start the dev server
```

- `react-ts` gives you React + TypeScript boilerplate, ready to go
- Open the printed `localhost` URL to see it live

Build Tools and Dev Servers

Two pieces of modern web tooling:

- **Build tool:** turns your source (many TS / JSX / CSS files) into optimized files a browser can run (`src/` → `dist/`)
 - bundled, minified, **tree-shaken** (unused code dropped), and obfuscated
- **Dev server:** runs your app locally while you develop, with **hot reload**: save a file and the change appears instantly - no manual refresh or rebuild

What Is Vite?

Vite is a modern tool that is **both** a build tool and a dev server.

- `yarn create vite` - scaffolds the project (your boilerplate)
- `yarn dev` - starts the dev server with hot reload
- `yarn build` - produces the production build (`src/` → `dist/`)

`src/` vs. `dist/`

- `src/` : your source code: the files you write (TypeScript, JSX, CSS)
- `dist/` : the build output: optimized, browser-ready files generated from `src/`
- You **edit** `src/` ; you **never hand-edit** `dist/` - it's regenerated each build
- The browser ultimately runs what's in `dist/`

Compilation and Building

Compilation turns your source into what browsers can actually run: TypeScript → JavaScript, JSX → JS - then it's bundled and minified.

```
yarn build          # compile src/ → dist/ once  
yarn build --watch  # rebuild automatically on every save
```

- `yarn build` runs the project's build script (Vite) to produce `dist/`
- `--watch` keeps watching your files and rebuilds as you change them

What Is Yarn? (vs. npm)

Both are **package managers** - they install and manage the libraries your project depends on (everything in `node_modules`).

- **npm** ships with Node; **Yarn** is a popular alternative - the one we use
- Both read `package.json` and write a lockfile for reproducible installs (`yarn.lock` vs. `package-lock.json`)

```
npm install           → yarn
npm install <pkg>     → yarn add <pkg>
npm run <script>      → yarn <script>
npm uninstall <pkg>  → yarn remove <pkg>
```

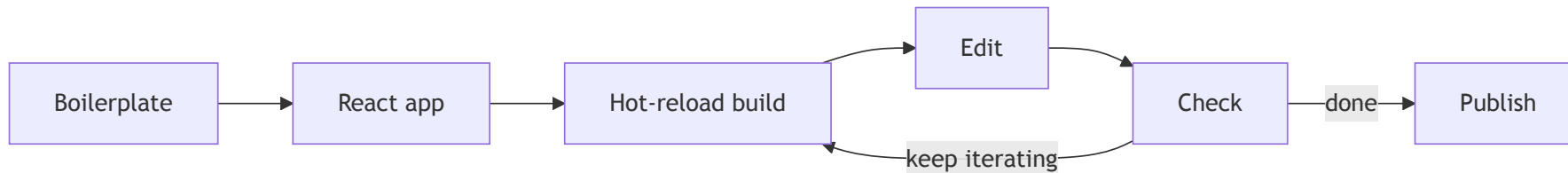
Why SPAs? (Reminder)

Traditional sites reload the **whole page** from the server on every click. A **single-page app (SPA)** loads once, then updates only the parts that change.

- **App-like feel**: no jarring full-page reloads
- **Less data** after the first load - fetch just the data, not whole pages
- **UI keeps its state** as you move around
- React is built for this: components re-render when their data changes

Pause and Check Your Mental Model

The development loop, start to finish:

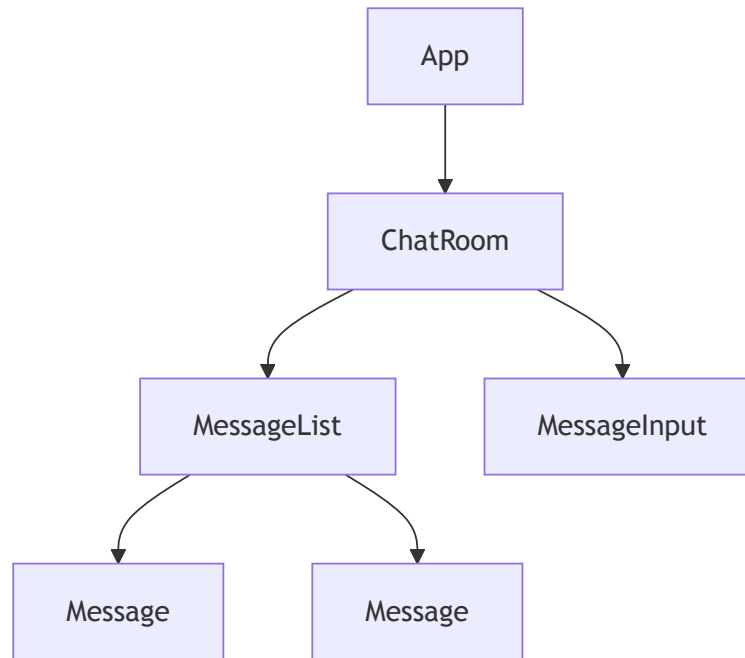


Components: The Mental Model

- A **component** is a function: **props in** → **UI (JSX) out**
- You build your interface as a **tree** of nested components
- They're **reusable building blocks** - think LEGO bricks
- A component can hold its own **state** and re-renders when that state changes

Composing Components

Break a screen into components, then nest them into a tree. **Props flow down** (parent → child); **events flow up** (child → parent via callbacks).



Code Together - Build a Fake Chatroom

Let's build a simple chat UI together - no real backend, messages just live in React **state**.

- **ChatRoom**: holds the list of messages in state
- **MessageList**: renders all the messages
- **Message**: displays a single message
- **MessageInput**: text box + send button that adds a message

"Fake" = messages are stored in state, not sent anywhere.

In-class reference: `demos/react-chat-app`